# AI Road Condition Intelligence & Maintenance System

Official Municipal Roadway Inspection & Prioritization Dossier

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| **Inspection Title:** | Live Test Upload | **Date:** | 2026-09-19 03:54 UTC |
| **Road Name:** | Test Highway | **Segment:** | Sector 4 Main Corridor |
| **Coordinates:** | 28.6139, 77.209 | **Priority:** | Medium |
| **Total Defects:** | 1 | **Peak Risk:** | 46.0/100 (Medium) |

## Detected Road Defects & Surface Hazards

| # | Defect Type | Confidence | Severity | Risk Score |
|---|---|---|---|---|
| 1 | Pothole | 82.0% | Medium | 46/100 |

## AI & Civil Engineering Maintenance Recommendations

**Source:** Rule-Based Fallback
**Urgency:** Within 7 Days
**Action:** Square edges using mechanical saw-cut, clean debris with compressed air, apply bitumen bonding emulsion, and tamp with polymer-modified asphalt compound.

**Explanation:** The defect was evaluated as 'Medium' severity with a risk index of 46/100 based on the detected pothole characteristics, an AI model detection confidence of 82.0%, and traffic exposure on a Medium importance roadway. The defect exhibits significant potential to compromise road safety and trigger progressive sub-base pavement deterioration if left unmitigated.